

A review on synchronization methods for grid-connected three-phase VSC under unbalanced and distorted conditions

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Keywords

<<Voltage source converter (VSC)>>, <<Three-phase system>>, <<Single-phase system>>, <<Signal processing>>, <<Power quality>>.

Abstract

The requirements for grid-connected voltage source converters (VSC) are increasing with the evolution towards smartgrids. In this context, synchronization methods for VSC have become a key feature for a fast and reliable control. The first synchronization methods, which were originally dedicated for communication applications have been gradually adapted to power systems applications and, today, an extensive collection of techniques exist. Although partial reviews and comparisons exist, a comprehensive classification of these approaches is needed. This paper presents an all-embracing survey of synchronization methods for grid-connected series and shunt VSC applications.

Introduction

The amount of power-electronics-based equipments and apparatus connected to the electrical transmission and distribution grid is progressively increasing as a consequence of the evolution towards smartgrids. Under unbalanced and highly distorted grid conditions, the control performance of grid-connected VSC apparatus depend upon a fast, accurate and reliable synchronization method. Therefore, the selection of the synchronization method constitutes a crucial element for guaranteeing a high quality operation. Synchronization methods must account for the following features:

- **Distortion rejection capability and noise immunity:** When harmonic pollution exists the synchronization method should be able to correctly filter grid-side harmonics in order to track the fundamental cleanly.
- **Frequency adaptivity:** While operating at weak grids system frequency might deviate from the nominal values. If grid frequency is susceptible of suffering from recursive variations, the selected synchronization method must be able to cope with those changes without losing synchronism.
- **Phase-angle adaptivity:** Some types of voltage-sags might generate sudden phase-jumps. The synchronization method must be able to correctly detect and ride-through these jumps.
- **Unbalance robustness:** At distribution level unbalances are often due to the amount of single-phase connected loads. At higher voltage levels, trains, for instance, are single-phase connected loads that generate temporary unbalances in the grid. Additionally, faults can also be source of transient unbalances.
- **Dynamics/convergence time:** The interest of a fast synchronization method is specially highlighted during transient events such as voltage sags or swells.
- **Structural simplicity:** Including simplicity of design, tuning and implementation.

- **Accuracy.**
- **Computational burden.** Kalman estimators, for example, offer good accuracy and performance but they are usually computationally heavy.

Grid-connected line-commutated converters for transmission systems used zero-crossing detection circuits. In spite of its simplicity, this method is limited by its dynamic performance, as zero-crossings are only detected every half-cycle of the grid frequency. For the last decades a myriad of solutions have been proposed in literature for grid-synchronization of VSCs and thus, the objective of this paper is to provide an overview (fig.1) of the main trends in synchronization methods. Probably the highest level classification divides synchronization methods into single-phase and three-phase structures. Single-phase VSC-based devices use single-phase structures, while three-phase VSC devices can either employ three single-phase units or one three-phase unit depending on the type of control [1, 2]. Three-phase VSC mostly use three-phase rather than single-phase structures.

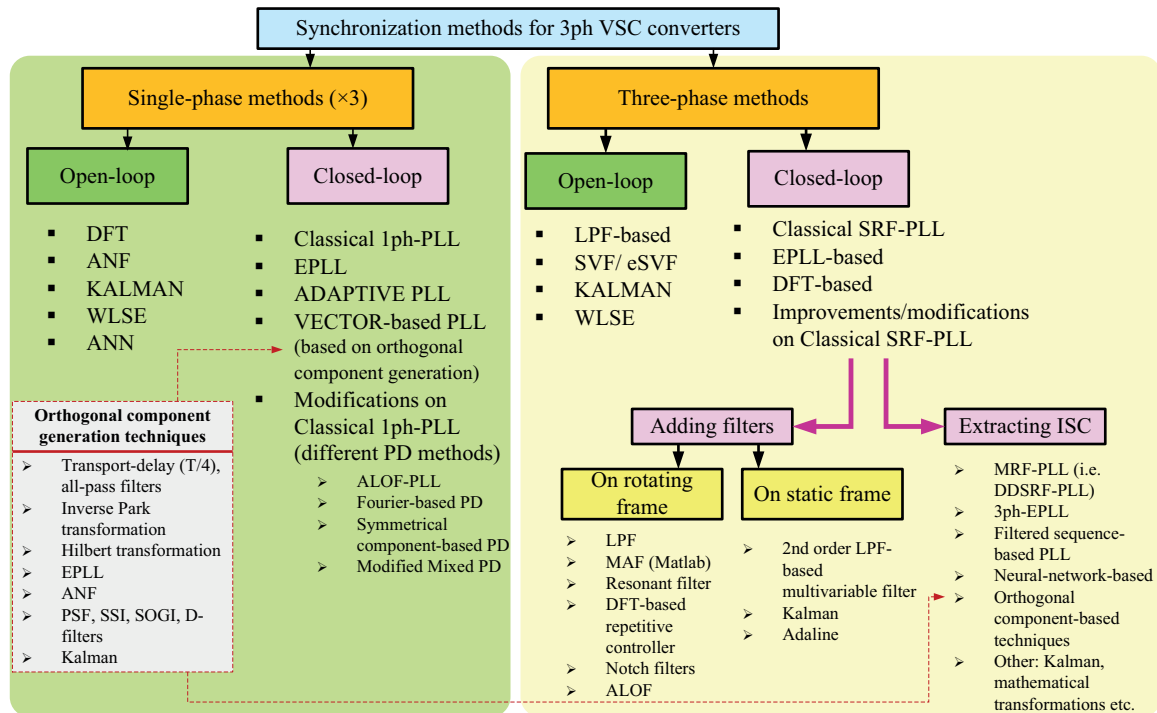


Figure 1: Classification of synchronization methods.

Single-phase synchronization systems

Single-phase synchronization methods can be categorized into open-loop and closed-loop methods. Open-loop methods directly estimate the magnitude, phase and frequency of the incoming signal whereas in closed-loop methods the estimation of the phase is updated adaptatively through a loop mechanism. This loop aims at locking the estimated value of the phase to its actual value [3].

Single-phase open-loop methods

Open-loop methods are often based on some type of filtering. Their performance depends on their capability of filtering distorted signals and their adaptability to system changes such as frequency and phase. The main approaches are based on discrete fourier transform (DFT), weighted-least-square-estimation (WLSE), adaptive notch filter (ANF), Kalman filtering and artificial neural networks (ANN).

DFT-based techniques [4] are appropriate for very distorted environments where good filtering characteristics are needed [5]. However, when the DFT sampling is asynchronous to the fundamental grid frequency (this is, when the grid frequency varies) phase errors are produced. Two solutions can be foreseen: (a) To correct the sampling window to match the grid period [6] or; (b) To add a phase offset to cancel the error produced by the recursive DFT algorithm [6, 5, 7]. According to [6] the window correction method achieves better performance than the phase-correction method because it ensures that harmonic components are rejected up to the aliasing frequency. However, not every systems allow working at variable sampling frequency. The presented compensation methods are able to cope with frequency change rates of up to 40 Hz/s.

ANF-based techniques [8, 9] provide instantaneous values of the various estimated frequency components in addition to the values of their frequencies, amplitudes and phase angles. Moreover, it does not require further burden than recursive DFT algorithm and it is not window-based. The proposed structure exhibits a longer transient time than some fast digital algorithms, such as those based on DFT or Kalman filtering.

Kalman filtering [4, 5, 10, 11] has shown to be a very accurate solution even under frequency variations. Nevertheless, it presents two main drawbacks: (a) the selection of covariance matrices and (b) the computational burden. Nevertheless, selection of gains and filter coefficients can be calculated offline.

WLSE [12] is presented as a fast and robust (insensitive) phase angle estimation algorithm that operates well even under sudden voltage conditions. Additionally, it estimates the value of the varying frequency.

ANN-based techniques based on adaptive linear combiner (Adaline) yield more accurate and faster estimations than Kalman filters [13]. The weight vector of the adaline generates the Fourier coefficients of the signal. This approach is highly adaptive and is able to follow nonstationary signals. Other neural-network techniques based on Hopfield type feedback have been proposed in [14].

Single-phase closed-loop methods

The first single-phase phase-locked loops (PLLs) were presented by Appleton and Bellescize as early as 1923 and 1932, respectively. For the late 1970's the theoretical foundations of PLLs were well established [15, 16] but they could only be comprehensively implemented with the development of integrated circuits (IC) [17].

Basic concept of single-phase PLL

A PLL is a device which controls the phase of its output in such a way that the phase error between the output phase and the reference phase is minimized. The block diagram of a PLL, shown in fig.2, is composed of the a phase-detector (PD), a loop-filter (LF) and a voltage-controlled oscillator (VCO). The phase difference between the input and the output is measured by the PD (the most intuitive form of a PD is a multiplier). Then, the output of the LF gives an error signal proportional to the phase difference between the input and output signals. Finally, the VCO generates the output signal depending on the phase error provided by the LF [17].

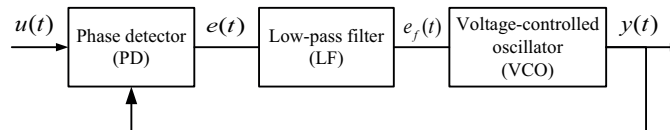


Figure 2: Basic topology of PLL.

The PD output is composed of a low-frequency and a high-frequency term. The LF, which is often a PI filter, is in charge of damping the high-frequency term but it does not totally eliminate it. The low-frequency term is a nonlinear function of the difference of the input and the output phase-angles. The magnitude of the PD output depends on the magnitude of the input signal and thus, voltage sags have an influence on the PD gain. Additionally, input harmonics on $x_i(t)$ may propagate through the feedback loop interfering on the control loop [18, 19].

Several improvements to these features have been proposed in the literature. In particular, alternative ways to construct the PD block have drawn much attention from the research community. Some of the most popular single-phase closed-loop approaches include:

- **Enhanced PLL (EPLL):** The EPLL approach presented in [20, 3] proposes a new PD which is able to remove the double-frequency ripple and provide an error-free estimate of the phase-angle and frequency when the input signal is a pure sinusoidal. This basic structure can be completed by additional blocks as described in [18], which makes this solution insensitive to harmonics and interharmonics. The EPLL seems an interesting solution for single and three-phase VSC synchronization.
- **Adaptive PLL:** The *adaptive PLL* presented in [19] demonstrates settling time and overshooting far below classical PLL systems. Testing with input harmonic content confirms robust responses and minimal harmonic propagation. The drawback of this structure is the lack of a generic control design approach.
- **Vector-based PLL:** Vector-based PLL is based on the three-phase synchronous-frame-PLL, which uses park transformation as PD. Unlike three-phase systems, single-phase systems do not naturally provide an $\alpha\beta$ static frame. For this reason, the single-phase input signal is considered as the α component while the β component needs to be generated in quadrature with the α components. The challenge consists in generating a signal 90 degrees phase-shifted from the input fundamental. This

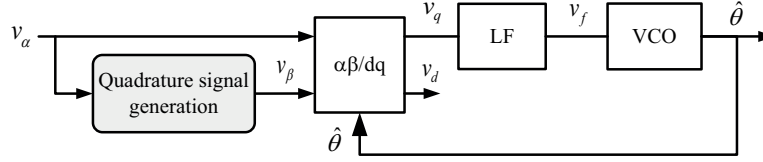


Figure 3: SRF-PLL for single-phase signals.

issue is addressed by many authors and there is plenty of alternative solutions: Transport-delay and all-pass-filters [21], Inverse-park transformation [22, 21], Hilbert transformation [21], EPLL [3], ANF [23], Stationary-frame Generalized Integrators (SGI) [24], Second order generalized integrators (SOGI) [25] and D-filters [26, 27], Kalman [28].

From the abovementioned methods, transport-delay, all-pass filters, inverse-park transformation and Hilbert transformation present frequency dependency. For this reason, they are not advisable in frequency variable environments. For the Kalman method errors due to frequency deviations can be alleviated either increasing the time constant or adjusting the algorithm sample time. SGI, SOGI and D-filters are based on similar concepts and their main characteristics are that (i) they filter the voltage signal without delay and (ii) they are frequency adaptative.

Three-phase synchronization systems

As in the single-phase case, three-phase synchronization methods can be broadly classified into open-loop and closed loop systems [3].

Three-phase open-loop synchronization methods

The following main open-loop methods have been identified.

- **Low-pass filtering (LPF) techniques:** Filtered signals are normalized and passed through a rotation matrix in order to compensate for the phase lag due to LPFs [29, 30]. A lower cut-off frequency guarantees a better filtering of the input signal at the cost of a slower response. The major drawback of the base solution, however, is its dependency with the grid-frequency (since the phase displacement depends on the center frequency) and its sensitivity to voltage unbalance [3]. Moreover, this method is unsuitable for applications where phase-jumps occur [29]. Different solutions have been proposed to overcome these disadvantages: a solution based on two frequency-adaptive sequential filters [31] and, an approach based on moving-average and predictive filters [32].
- **Space-vector filter (SVF) techniques:** Although the basic SVF [29] behaves well face to phase-jumps and harmonic distortion, it introduces a phase-shift when the grid-frequency varies. The extended SVF (eSVF), also presented in [29], is designed to be frequency adaptive. Nevertheless, it seems that tuning of the added PI-regulator is in conflict with frequency tracking and voltage harmonics sensitivity.
- **Kalman filtering (KF) techniques:** Kalman techniques have successfully been implemented in [4, 29, 33, 10, 11, 5]. These synchronization techniques can work in distorted and unbalanced environments, they can cope with phase jumps and they are frequency adaptative. Their main drawback is their computational burden, their higher convergence time [5] and the difficulty in selecting the optimal weighting matrices (process and measure covariance matrices) [29].
- **Weighted Least Squares Estimation (WLES) techniques:** The proposed angle detection algorithm acts without delay when a sudden voltage sag or unbalance occurs, it estimates positive and negative sequences separately, and accomodates grid frequency variations [34, 3]. This method can be distinguished from conventional filtering techniques in its fast transient response. However, [3] reports long transient intervals in detecting frequency changes, computational problems related to least-squares methods and sensitivity to noise and distortions.

Three-phase closed-loop synchronization methods

Closed-loop methods operate in a closed-loop structure which regulates an error signal to zero. Probably the most well-known and spread closed-loop synchronization method is the synchronous rotating frame PLL (SRF-PLL), which became popular in the late 90's [35, 36, 22]. Similar approaches to the SRF-PLL, and with equivalent shortcomings, are the pq-PLL [37, 38], which can be easily interpreted thanks to the instantaneous real and imaginary power theory and the orthogonality-based PLL [39].

In [40], an alternative PD method based on DFT and oriented towards aircraft applications is proposed and in [41, 42] two adaptations of the single-phase EPLL for three-phase systems are introduced. Unique features of such filters are frequency adaptivity, unbalance mitigation and structural simplicity/robustness.

Classical three-phase PLL or SRF-PLL

Analogous to the single-phase PLL, and as depicted in fig. 4, the three-phase SRF-PLL can be divided in a PD block that is constituted by a park-transformation, a LF which is often a PI-regulator and a VCO that is usually an integrator. The regulator is in charge of setting the direct (or the quadrature) components of the input grid voltage to zero and the choice of the feedback gains requires an small-signal analysis [35, 36, 22]. Tuning techniques based on the Wiener Method (based on stochastic characteristics of noise) [36] and Symmetrical Optimum [35] have been proposed.

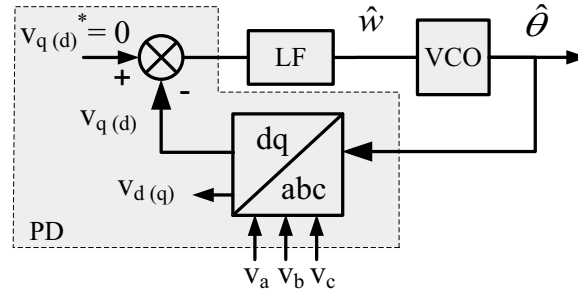


Figure 4: Schematic representation of three-phase SRF-PLL.

The dynamic performance of the filter must satisfy fast tracking as well as good filtering characteristics. However, the SRF-PLL cannot satisfy both requirements simultaneously. A low dynamics filter produces a very damped and stable output at the cost of a longer synchronization time. A fast tuned filter synchronizes quickly to the grid voltage but the distortions of the input signal pass through the control loop and they are reflected in the output. A trade-off is necessary when designing the control parameters of the SRF-PLL.

Unbalanced voltages generate a superposed second-order harmonic in dq components which is very near from the fundamental frequency. This issue can be addressed by lowering the cut-off frequency of the loop filter at the expense of lowering the time response of the SRF-PLL. A possible solution, which will be addressed later, is extracting the positive sequence of the input voltage for locking the PLL to it. Another robust solution, based on compensating for the frequency and phase-angle distortion is proposed in [43].

Filters in SRF-PLL

Since the effect of unbalance and harmonic distortion propagates towards the static- and rotating-frame voltage components, it is possible to damp these components by an additional filter placed before the LF. This filter can be located in the rotating-frame (fig.5(a)) or in the static-frame (fig.5(b)).

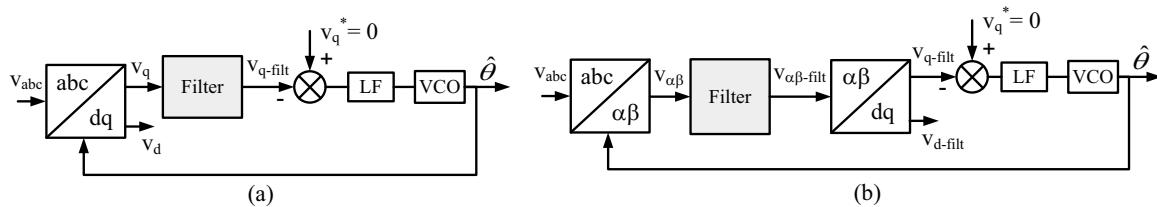


Figure 5: Filters in SRF-PLL loops: (a) in the rotating-frame (b) in the static-frame.

In the rotating frame, solutions based on LPFs [44, 45, 46], moving average filters (MAFs) [47, 45, 48] (used in the standard PLL block of Matlab SimPowerSystems toolbox), resonant filters [44, 49, 45], DFT-based repetitive controllers [50, 45], notch-filters [44] and adaptive linear optimal filters (ALOF) [51], among others, have been used.

LPF-based techniques are simple to implement but they present two shortcomings: (a) the narrower the bandwidth of the filter is, the better the immunity to distortion is at the cost of slower transient times and (b) it is frequency-dependent introducing variable phase-shifts in the filtered signals.

On the contrary to LPFs, **resonant filters** as such used in [49] do not introduce any phase-shifts at the resonant frequency and it offers a superior harmonics rejection capability. Notch-filters filter the second-harmonic component but their response time is slow [44].

MAF-based techniques demonstrate excellent second-harmonic cancellation and superb elimination of external AC voltage harmonics but their transient response is unfavourable (due to the length of the moving window) [19]. The poor phase angle tracking under reduced voltages seems to be solved with an automatic gain block in the Matlab/SimPowerSystems blockset.

The **repetitive controller** improves the rejection capability of the PI controller by amplifying the second harmonic. It works essentially like a bandpass filter in which the odd harmonics are filtered while the even harmonics are not. This way, the proportional gain of the PI controller is indirectly increased and thus, the rejection capability. This controller is implemented by means of a DFT algorithm and is robust against frequency variations and phase-jumps.

The **ALOF** uses a least-mean-square algorithm that looks like an adaptive linear neural network (ADALINE) algorithm. The ALOF shows the characteristics of a band-pass filter at fundamental frequency and a notch filter at harmonic frequencies and it shows a good tracking accuracy, dynamic response and immunity to grid voltage disturbances [51].

In the static-frame, solutions based on a second-order LPF-based multivariable filter, a Kalman filter and a ADALINE algorithm have been presented in [52]. The Kalman filter and the ADALINE based filter give similar results, which are better than the second-order LPF, but ADALINE filter is specially interesting for its simplicity.

Symmetrical sequence extracting techniques applied to SRF-PLL

An interesting solution, other than filtering the looped input (e.g. the quadrature component), is to extract the positive sequence from the grid-voltage and feeding this sequence to the classical SRF-PLL as illustrated in fig.6. This approach allows having a distortion-free input at the PLL that will, in turn, enable tuning the PLL with a higher bandwidth. Many different alternatives have been proposed for extracting the instantaneous symmetrical components (ISC) online. The following points overview some of them:

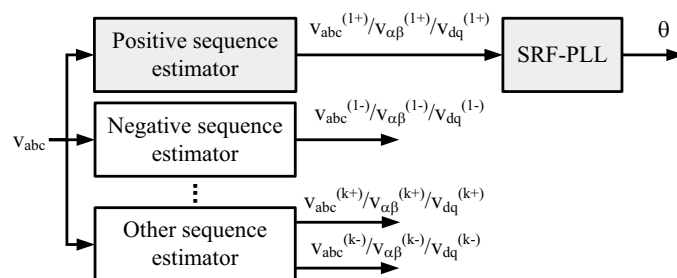


Figure 6: Sequence extraction and use of fundamental positive sequence with the SRF-PLL.

- **Multiple reference frame based PLL (MRF-PLL).** The use of multiple reference frames allow extracting the ISC separately. Two similar approaches can be observed: the decoupled double synchronous reference frame PLL (DDSRF-PLL) [46, 53] and the multiple reference frames of [54]. These methods extract positive and negative sequences by means of two reference frames rotating at the same angular speed in the positive and in the negative direction. The couplings between rotating axes are removed by decoupling networks and the magnitudes of the sequences are obtained by LPFs. These methods give excellent results in unbalanced networks but are not as performant face to strong harmonic distortion due to the bandwidth reduction necessary for obtaining satisfactory results [55].
- **Three-phase EPLL.** In [42, 56] an EPLL-based method is proposed for online calculation of ISC. The suggested technique is mathematically derived based on an optimization problem and is able to estimate the magnitudes, phase-angles and frequency of ISC. This method exhibits longer transient time than some fast digital algorithms such as those based on FFT. This constitutes a limitation for those applications which require very fast transient period. On the other hand, it has a relatively large structure which adds additional complexity. However, the authors provide design guidelines to reduce its complexity and implementation cost. The proposed solution is specially adapted for those applications which require accurate estimation of parameters in unbalanced conditions.
- **Filtered-sequence based PLL.** [48] proposes a variable-frequency grid-sequence detector based on a quasi-ideal low-pass filter stage and a PLL. The structure includes the use of Park transformations (Park and inverse-Park) and moving average filters. This solutions has shown a very good performance face to strong grid conditions showing a remarkable advantage in comparison with

the newest and more sophisticated positive-sequence detectors: its simplicity. [31] proposes a positive sequence extraction technique based on sequential LPF filters.

- **Neural-network-based PLL.** Adaptive linear neural-network extractors (ADALINE) show easy synthesis and programming, adaptive capability to the change in grid voltage and fast response [57, 58].
- **Orthogonal component-based techniques.** Symmetrical sequence components can be extracted based on orthogonal components of the input voltages in static or natural frames [59, 60, 3]. This method is largely used and many approaches exist. The difficulty of the positive-/negative sequence calculator (PNSC) relies on the ability of calculating quadrature signals. Many different approaches have been evaluated in literature. Most of them have already been reviewed in the single-phase *vector-based PLL* section but they are briefly mentioned hereafter: all-pass filters [61], one-fourth of a period delayed signal used in Delayed Signal Cancellation method (DSC) [62, 44], EPLLs [3, 59], ANFs [23], dual second order generators (DSOGI) [59, 60], PSF/SSI [24, 55], Kalman filters [11].

In their basic concept all-pass filters and DSC are grid frequency dependent, but [62], for example, gives a solution to compensate for errors produced by variable frequency in DSC. EPLL and ANF methods are nonlinear adaptive filters that show a very good degree of immunity and frequency adaptivity. Their only drawback [59] is that they are implemented in the natural frame and thus, (a) they require more computational burden than those methods implemented in the static-frame and (b) zero-sequence components are not blocked in Clarke transformation. According to [55] DSOGI is the most performant among DSOGI, PSF and SSI methods. Moreover, SSI does not perform well in presence of unbalances. Kalman filters show a good accuracy in comparison with EPLL [3], but they are usually computationally heavy.

- **Kalman.** [63] proposes an ISC extracting technique based on complex Kalman filter.

Conclusions

Since the advent of the first PLL in 1923 many different approaches have been successively proposed in order to improve preceeding ones. Today, even if new publications continue to arise, it can be said that synchronization methods for VSC converters is a pretty mature subject considering the amount of existing publications.

This work provides a clear outline of all the existing approaches considering that, until now, only partial reviews existed. This survey overviews and classifies a wide assortment of publications but, unfortunately, and due to lack of space, deeper descriptions could not be given. The authors believe that a second part of this review describing the main techniques in a more didactic way could be helpful for the research community.

On the other hand, many publications make comparisons among several techniques but, the lack of harmonisation in the testbenches does not make possible the comparison between different publications. It could be interesting to fix common standards and test-benches to compare the characteristics of synchronization methods in a rigorous way.

Synchronization methods can be evaluated according to dynamics/convergence time, accuracy, distortion/disturbance rejection, phase-angle adaptivity, frequency adaptivity, unbalance robustness, noise immunity, structural simplicity (design, tuning and implementation), computational burden and single or three-phase utilization. It is up to users to choose the method that suits them better.

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